

Lightweight Multimedia Anomaly and Integrity Detection for Consumer IoT Using Knowledge Distillation

Fasee Ullah^{ID}, Hamid Asmat, Arfat Ahmad Khan^{ID}, Muhammad Ismail Mohmand^{ID}, Farman Ali^{ID},
Rayan Hamza Alsisi^{ID}, Theyazn H. H. Aldhyani^{ID}, and Daehan Kwak^{ID}, *Senior Member, IEEE*

Abstract—The rapid growth of Consumer Internet of Things (CIoT) devices has significantly increased real-time multimedia data exchange, heightening vulnerability to attacks targeting audio, video, and image content. This paper introduces Multimedia Anomaly and Integrity Detection using Knowledge Distillation (MAID-KD), a lightweight multi-task framework that performs anomaly detection and integrity verification in CIoT environments. MAID-KD leverages a Transformer-based teacher model to extract rich spatio-temporal features from multimedia streams, while a compact CNN-LSTM student model optimized for edge deployment is trained through feature alignment, soft-target distillation, and variational projection. Experimental results demonstrate that MAID-KD achieves superior accuracy and F1-score compared to state-of-the-art baselines, while reducing model size and inference latency by over 60%. These results highlight MAID-KD's ability to deliver scalable, privacy-preserving, and multimedia-aware security for CIoT devices such as smart surveillance systems, health-monitoring wearables, and connected home platforms.

Index Terms—Multimedia security, malware detection, intrusion detection system (IDS), CIoT, edge computing, knowledge distillation, CNN-LSTM hybrid, privacy preservation, real-time threat detection.

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Fasee Ullah is with the Department of Computing, Universiti Teknologi PETRONAS, Seri Iskandar 32610, Malaysia (e-mail: fasee.ullah@utp.edu.my).

Hamid Asmat is with the Department of Computer Science, University of Haripur, Haripur 22620, Pakistan (e-mail: hamid@uoh.edu.pk).

Arfat Ahmad Khan is with the Department of Computer Science, Khon Kaen University, Khon Kaen 40002, Thailand (e-mail: arfatkhan@kku.ac.th).

Muhammad Ismail Mohmand is with the Department of Computer Engineering and the Faculty of Engineering and Natural Sciences, Istanbul Atlas University, 34408 Istanbul, Türkiye (e-mail: muhammad.mohmand@atlas.edu.tr).

Farman Ali is with the Department of Applied AI, Sungkyunkwan University, Seoul 03063, South Korea (e-mail: farman0977@skku.edu).

Rayan Hamza Alsisi is with the Department of Electrical Engineering, Islamic University of Madinah, Madinah 41411, Saudi Arabia (e-mail: ralsisi@iu.edu.sa).

Theyazn H. H. Aldhyani is with the Applied College, King Faisal University, Al-Ahsa 31982, Saudi Arabia (e-mail: taldhyani@kfu.edu.sa).

Daehan Kwak is with the Department of Computer Science and Technology, Kean University, Union, NJ 07083 USA (e-mail: dkwak@kean.edu).

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I. INTRODUCTION

THE rapid expansion of Consumer Internet of Things (CIoT) ecosystems, including smart surveillance cameras, voice-enabled home assistants, health-monitoring wearables, and connected vehicles, has dramatically increased the volume of multimedia data (e.g., video, audio, and images) transmitted across heterogeneous consumer devices. These multimedia streams carry high-value information and are susceptible to attacks such as video feed tampering, deepfake audio injection, data exfiltration, and unauthorized redistribution, posing severe threats to user privacy and trust. As the adoption of CIoT devices accelerates [1], [2], ensuring robust security and privacy has become increasingly challenging. These devices operate under strict computational and energy constraints, making them attractive targets for sophisticated attacks, including multimedia-specific exploits such as frame injection, video/audio stream tampering, and steganography-based data hiding [3], [4].

Lightweight anomaly detection solutions have emerged as a primary defense for CIoT environments; however, deploying conventional deep learning models at the edge remains challenging due to energy, latency, and memory limitations [5]. This is particularly critical for multimedia applications such as live surveillance and smart home monitoring, where cloud offloading is often infeasible. Complementary multimedia security mechanisms such as encrypted data transmission, steganography detection, and watermarking-based integrity checks have been proposed, but integrating these into edge frameworks is still in its early stages [4], [6]. Multimedia streams often contain sensitive user information and can be manipulated to bypass authentication systems or mislead automated surveillance. Existing solutions, like watermarking and multimedia encryption, provide partial protections but remain difficult to integrate with lightweight anomaly detection models. This gap highlights the need for unified approaches that secure both the device and its multimedia content without sacrificing real-time performance.

Knowledge Distillation (KD) has become a promising strategy to enable lightweight models by transferring knowledge from large teacher networks to compact student models [7]. In the context of CIoT, KD enables on-device anomaly detection and multimedia integrity verification without transmitting raw multimedia data externally, thereby

reducing privacy risks. However, hybrid frameworks that combine KD with multimedia encryption or federated learning still face issues like task interference and model inversion attacks, particularly when teacher models are trained on multiple detection objectives simultaneously [8], [9], [10].

A key limitation of existing approaches is the isolated treatment of anomaly detection and integrity verification as separate tasks, which hinders their ability to detect blended or multi-stage attacks where abnormal data patterns overlap with tampered multimedia content [11]. Single-task solutions also struggle to adapt across heterogeneous consumer platforms where multimedia data types and attack surfaces vary significantly [12]. These gaps underscore the need for a unified, multi-task approach tailored for consumer-grade devices that also protects multimedia data integrity.

The core research problem addressed in this work is how to design a lightweight and privacy-preserving framework capable of jointly performing multimedia anomaly detection and integrity verification under the computational and bandwidth constraints of consumer IoT environments. To address these challenges, this study proposes a Multimedia Anomaly and Integrity Detection using Knowledge Distillation (MAID-KD) framework that consolidates anomaly detection and multimedia integrity verification in a single student model optimized for CIoT devices. The teacher model leverages a Transformer-based encoder to capture long-range dependencies from multimedia-rich data, while the student employs a CNN-LSTM hybrid architecture to preserve spatio-temporal representations in a lightweight form. Knowledge transfer is performed using teacher-guided feature alignment and probabilistic variational projection, with a multi-objective optimization function balancing classification, reconstruction, and distillation losses. This design enables MAID-KD to provide strong detection accuracy while maintaining minimal computational and memory overhead, making it suitable for real-time deployments in multimedia-intensive CIoT environments such as smart surveillance systems, connected home cameras, and health-monitoring wearables.

The proposed MAID-KD framework focuses on multimedia-centric security, unlike existing KD-based IoT solutions such as FLEKD [13], and DTKD-IDS [14], which primarily rely on network traffic data. MAID-KD introduces a unified multi-task distillation system that performs both anomaly detection and multimedia integrity verification within a single lightweight model. It further incorporates a variational knowledge projection module to handle uncertainty in multimedia data, enabling the student model to detect blended or tampered content efficiently under resource constraints. These design choices distinguish MAID-KD as a novel, real-time solution tailored for consumer IoT environments.

The contributions of this work are as follows:

- A unified anomaly detection and multimedia integrity verification framework using task-aware knowledge distillation.
- A training pipeline that prevents raw multimedia data exposure while enabling flexible deployment on consumer hardware.

- A detailed evaluation of accuracy, adaptability, multimedia data integrity, and privacy across multiple multimedia attack scenarios.

The remainder of this paper is organized as follows: Section II reviews related work on the use of KD in secure IoT settings. Section III details the proposed MAID-KD framework and its underlying algorithms. Section IV presents the experimental setup. Results are explained in Section V. Finally, Section VI concludes the paper.

II. RELATED WORK

Security and privacy in Consumer Internet of Things (CIoT) environments have become increasingly critical with the rise of multimedia data exchange through devices such as wearables, smart appliances, and connected vehicles. Early studies emphasized securing multimedia content in consumer electronics using cryptographic and biometric mechanisms [15], [16]. These foundational methods focused on protecting audio-visual data streams and consumer privacy across home networks. More recently, multimedia-centric threats, such as data tampering and covert exfiltration, have emerged in CIoT environments, motivating stronger protection for user identity, behavior, and content integrity [17], [18]. Recent work has proposed specialized tampering detection frameworks for images and video using multiscale attention and spatio-temporal analysis [19], [20], as well as deep learning-based methods for audio deepfake detection [21]. These advances are essential to preserve the authenticity of multimedia data generated by CIoT devices.

Blockchain and federated learning (FL) are increasingly adopted for distributed multimedia security. For instance, redactable blockchain frameworks, such as TMRB, address immutability and content redaction needs in multimedia scenarios [22]. FL integrated with blockchain has been shown to improve trust and privacy by eliminating the need for central data aggregation [23], [24]. Similarly, Zero-Trust Architecture (ZTA) approaches leverage AI-driven profiling to verify interactions in real-time [14], [25], offering a robust framework for hostile environments where multimedia traffic must be authenticated continuously. Complementary watermarking techniques, such as secure watermarking schemes tailored for IoT devices [26], [27], have also been employed to protect ownership and detect content manipulation in multimedia streams.

KD is widely explored for enabling lightweight edge models capable of processing multimedia data streams locally. Multi-branch KD frameworks have been shown to perform intrusion detection and malware analysis without cloud dependency [28], [29]. Unified KD-based models capable of handling multiple detection objectives simultaneously are of particular relevance to multimedia-driven CIoT, where blended attacks are common. Recent reviews also emphasize the importance of maintaining content integrity during KD training for regulatory and consumer trust [30].

Agent-based and situationally aware Intrusion Detection Systems (IDS) have also emerged as promising solutions for multimedia environments. For example, privacy-preserving agent IDSs adapt dynamically using contextual multimedia

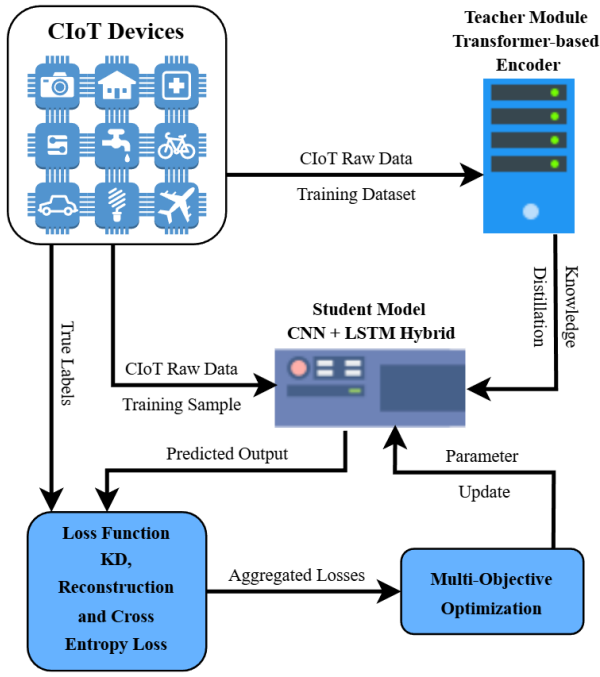


Fig. 1. Training Phase: The teacher model guides the student using knowledge distillation, while a multi-objective loss function optimizes the student parameters.

features [31], while deep feature extraction and compression have been proposed to reduce resource consumption on edge devices [32], [33], [34]. Additionally, cryptographic compression and watermarking techniques have been integrated with GANs and hybrid storage schemes to enable secure multimedia model transmission across CIoT networks [35], [36]. These advancements collectively highlight the evolving nature of multimedia security challenges and the growing focus on integrating lightweight, multi-task learning models for real-time protection in CIoT applications.

Existing methods are useful in certain domains. However, they are insufficient for real-time multimedia security in CIoT. Federated learning and blockchain provides high latency and communication overhead, which makes them unsuitable for low-power edge devices. The main difference between KD-based intrusion detection systems and the watermarking and encryption techniques is that network traffic is the primary target of intrusion detection, and the integrity of multimedia content is not addressed, while watermarking and encryption tools guarantee authenticity but do not detect anomalies in the traffic. Such limitations support the importance of having a single, lightweight, and privacy-aware solution that combines anomaly detection and multimedia integrity checking of resource-constrained CIoT devices.

III. THE PROPOSED MODEL

A. Overview of the Proposed MAID-KD Framework

The MAID-KD framework enhances multimedia security in CIoT devices by enabling on-device threat detection under resource constraints. As shown in Fig. 1 and Fig. 2, MAID-KD has two phases. In the Training Phase, heterogeneous multimedia IoT data (video, audio, images, telemetry) are

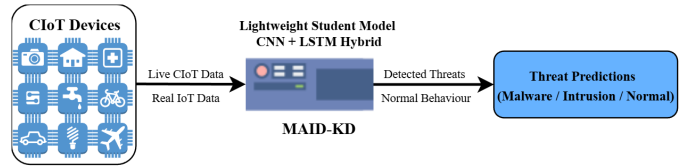


Fig. 2. Deployment Phase: The trained student model (MAID-KD) performs real-time threat predictions using live IoT data.

processed by a Transformer-based teacher model to learn rich spatio-temporal and cross-modal representations, while a lightweight CNN+LSTM student model is trained in parallel on smaller data batches. Knowledge is distilled through high-level features and soft targets, limiting the student's exposure to raw data and reducing privacy risks. In the Deployment Phase, the trained student runs locally on CIoT edge devices to analyze live multimedia streams, performing anomaly detection and anomaly type classification without teacher support or cloud dependency. This approach minimizes computational and bandwidth overhead while maintaining multimedia data confidentiality and integrity across CIoT environments.

B. System Model

We consider a distributed CIoT network comprising N edge nodes, $\mathcal{D} = \{d_1, d_2, \dots, d_N\}$, where each node processes multimedia data (e.g., video, audio, telemetry) under strict resource constraints. The framework combines a high-capacity teacher model T_ϕ , trained offline on large-scale CIoT datasets, with a lightweight student model S_θ deployed on edge devices. The objective is to transfer knowledge from T_ϕ to S_θ for anomaly detection and anomaly type classification, while maintaining user data privacy.

Each node d_i observes input-label pairs

$$\mathcal{Z}_i = \{(x_i^{(j)}, y_i^{(j)})\}_{j=1}^{n_i}, \quad (1)$$

where $x_i^{(j)} \in \mathbb{R}^d$ represents multimedia feature vectors and $y_i^{(j)} \in \mathcal{C}$ denotes the corresponding labels.

The global task set is defined as

$$\mathcal{T} = \{\mathcal{T}_{\text{anom}}, \mathcal{T}_{\text{type}}\}, \quad (2)$$

covering both anomaly detection and anomaly type classification. By distilling T_ϕ 's knowledge into S_θ across $\mathcal{T}$ in (2), the framework provides robust protection against multimedia-driven blended attacks in CIoT environments.

C. Teacher-Guided Feature Transformation

Let the input $x \in \mathbb{R}^d$ represent multimedia data such as video frames, audio streams, and telemetry packets commonly found in CIoT environments. These heterogeneous inputs are drawn from a distribution $\mathcal{X}$ and often include blended attack patterns, making robust representation learning essential. To address this, we define the teacher model $T_\phi : \mathbb{R}^d \rightarrow \mathcal{H}_T$ and the student model $S_\theta : \mathbb{R}^d \rightarrow \mathcal{H}_S$ as parameterized mappings that transform the input x into their respective latent spaces $\mathcal{H}_T$ and $\mathcal{H}_S$.

The teacher generates a high-level embedding of the input using a linear projection followed by a non-linear activation, as shown in Equation (3)

$$\mathcal{F}_T(x) = \sigma(\mathbf{W}_T x + \mathbf{b}_T), \quad (3)$$

where $\mathbf{W}_T \in \mathbb{R}^{k \times d}$ and $\mathbf{b}_T \in \mathbb{R}^k$ denote the trainable weights and biases. Analogously, the student performs a similar transformation into its latent space shown in Equation (4)

$$\mathcal{F}_S(x) = \sigma(\mathbf{W}_S x + \mathbf{b}_S), \quad (4)$$

where $\mathbf{W}_S \in \mathbb{R}^{k \times d}$ and $\mathbf{b}_S \in \mathbb{R}^k$ are the student parameters.

To ensure the student embedding space remains consistent with the teacher, we minimize the squared ℓ_2 -norm distance between $\mathcal{F}_T(x)$ and $\mathcal{F}_S(x)$ as defined in Equation (5)

$$\Delta_{\mathcal{F}} = \|\mathcal{F}_T(x) - \mathcal{F}_S(x)\|_2^2. \quad (5)$$

This objective is averaged across the data distribution, as in Equation (6), to obtain the empirical alignment loss

$$\mathcal{J}_1(\theta) = \mathbb{E}_{x \sim \mathcal{X}} [\Delta_{\mathcal{F}}]. \quad (6)$$

To further improve generalization, a regularization term based on the Frobenius norm of the student's weight matrix is added to Equation (7)

$$\mathcal{R}_1(\theta) = \lambda_1 \cdot \mathcal{J}_1(\theta) + \alpha \|\mathbf{W}_S\|_F^2, \quad (7)$$

where λ_1 balances the alignment term and α penalizes large weights.

In addition to latent feature alignment, the student also mimics the teacher's output distribution using temperature-scaled softmax logits in Equations (8) and (9)

$$\hat{y}_T = \text{softmax}\left(\frac{T_{\phi}(x)}{\tau}\right), \quad (8)$$

$$\hat{y}_S = \text{softmax}\left(\frac{S_{\theta}(x)}{\tau}\right), \quad (9)$$

where $\tau > 0$ is the temperature parameter that softens class probabilities. The divergence between the softened outputs is measured using the scaled Kullback–Leibler divergence Equation (10)

$$\mathcal{L}_{\text{KD}} = \tau^2 \cdot \text{KL}(\hat{y}_T \parallel \hat{y}_S). \quad (10)$$

Together, the feature alignment loss $\mathcal{R}_1(\theta)$ from Eq. (7) and the knowledge distillation loss $\mathcal{L}_{\text{KD}}$ from Eq. (10) establish a dual-objective function that enforces structural similarity in the latent space and consistency at the decision level. Algorithm 1 outlines the teacher-guided feature transformation process. It aligns the student's lightweight multimedia representations with the teacher's richer embeddings by minimizing feature-level and output-level discrepancies. The algorithm iteratively updates student parameters using feature alignment loss, KL-divergence on softened probabilities, and regularization, resulting in a compact yet accurate student model.

Algorithm 1 Teacher-Guided Feature Transformation

Result: Updated student parameters ω_S
Input: Multimedia input set $\mathcal{X} = \{x_i\}_{i=1}^N$, temperature τ , learning rate η , teacher weights ω_T , initial student weights ω_S , penalties λ, α
Output: Optimized student parameters ω_S
Initialize student weights $\omega_S := [\mathbf{M}, \mathbf{c}]$;
// Projection layer parameters
Set total loss $\mathcal{L} := 0$; // Initialize loss accumulator
foreach sample x_i in $\mathcal{X}$ **do**
 Compute teacher embedding: $\mathbf{z}_T := \sigma(\mathbf{A}x_i + \mathbf{a})$;
 // Rich multimedia features
 Compute student embedding: $\mathbf{z}_S := \sigma(\mathbf{M}x_i + \mathbf{c})$;
 // Lightweight representation
 for $j := 1$ **to** k **do**
 Feature alignment error: $\delta_j := (z_{T,j} - z_{S,j})^2$
 Accumulate feature loss: $\mathcal{L} := \mathcal{L} + \delta_j$
 end
 Regularization: $\mathcal{R} := \sum_{p,q} M_{pq}^2$
 Teacher and student logits:
 $\ell_T := T(x_i), \ell_S := S(x_i)$ Softened probabilities:
 $\pi_T := \text{softmax}(\ell_T/\tau), \pi_S := \text{softmax}(\ell_S/\tau)$
 KL divergence: $D := \sum_c \pi_{T,c} \log \frac{\pi_{T,c}}{\pi_{S,c}}$
 Total loss: $\mathcal{L} := \lambda \cdot \mathcal{L} + \alpha \cdot \mathcal{R} + \tau^2 \cdot D$
end
for $t := 1$ **to** $T_{\max}$ **do**
 Gradient: $\nabla \omega_S := \nabla_{\omega_S} \mathcal{L}$;
 // Backpropagation
 Parameter update: $\omega_S := \omega_S - \eta \cdot \nabla \omega_S$
end
return ω_S ; // Final trained student network

D. Variational Knowledge Projection

In multimedia-centric CIoT environments, data streams such as video frames, audio signals, and telemetry often contain overlapping attack signatures. To enhance the robustness of the student model against such blended anomalies, the extracted features are projected into a probabilistic latent subspace using variational inference. This enables the student model to learn compact multimedia-aware representations while maintaining uncertainty estimates for better generalization and resilience against adversarial noise.

We define a variational posterior distribution $q_{\theta}(z|x)$ over the latent variable $z \in \mathbb{R}^m$ conditioned on the input sample $x \in \mathbb{R}^d$. As expressed in Equation 11, this posterior is modeled as a multivariate Gaussian distribution

$$q_{\theta}(z|x) = \mathcal{N}(z \mid \mu_{\theta}(x), \Sigma_{\theta}(x)) \quad (11)$$

where the mean vector $\mu_{\theta}(x)$ and covariance matrix $\Sigma_{\theta}(x)$ are derived from the student feature space $\mathcal{F}_S(x)$. The mean mapping is computed as

$$\mu_{\theta}(x) = \mathbf{W}_{\mu} \mathcal{F}_S(x) + \mathbf{b}_{\mu} \quad (12)$$

where $\mathbf{W}_\mu \in \mathbb{R}^{m \times k}$ and $\mathbf{b}_\mu \in \mathbb{R}^m$ are learnable parameters. Similarly, the covariance in Equation 13 is obtained using an exponential activation to enforce positivity

$$\Sigma_\theta(x) = \text{diag}(\exp(\mathbf{W}_\sigma \mathcal{F}_S(x) + \mathbf{b}_\sigma)) \quad (13)$$

ensuring numerical stability and a valid Gaussian distribution.

Latent variable sampling follows Equation 14, where the reparameterization trick introduces stochasticity while preserving differentiability

$$z = \mu_\theta(x) + \Sigma_\theta^{1/2}(x) \odot \epsilon, \quad \epsilon \sim \mathcal{N}(0, \mathbf{I}) \quad (14)$$

This sampled latent vector z is subsequently decoded to reconstruct the original feature embedding using Equation 15

$$\mathcal{F}_z = \text{ReLU}(\mathbf{W}_z z + \mathbf{b}_z) \quad (15)$$

where $\mathbf{W}_z \in \mathbb{R}^{k \times m}$ and $\mathbf{b}_z \in \mathbb{R}^k$ parameterize the decoder.

The objective function in Equation 16 is designed to balance the reconstruction fidelity of multimedia features and the regularity of the latent space

$$\mathcal{L}_{\text{VAE}} = \mathbb{E}_{z \sim q_\theta(z|x)} [\|\mathcal{F}_S(x) - \mathcal{F}_z\|^2] + \beta \cdot \text{KL}(q_\theta(z|x) \| p(z)) \quad (16)$$

This loss combines two objectives: the first term preserves the accuracy of reconstructed multimedia features, while the second term regularizes the latent space to maintain generalization and prevent overfitting. The first term ensures that reconstructed features accurately reflect multimedia-specific information, while the second term enforces proximity to a prior $p(z) = \mathcal{N}(0, \mathbf{I})$ as defined in Eq. 17:

$$p(z) = \mathcal{N}(0, \mathbf{I}) \quad (17)$$

Regularization through this KL-divergence prevents overfitting and guarantees a well-structured latent space, which is critical for detecting subtle anomalies in compressed multimedia signals. This standard normal prior serves as a reference distribution that stabilizes the latent representation and avoids bias toward specific multimedia samples.

Finally, the scaled variational loss in Equation 18 is integrated into the overall optimization objective of the student

$$\mathcal{J}_2(\theta) = \lambda_2 \cdot \mathcal{L}_{\text{VAE}} \quad (18)$$

The coefficient λ_2 adjusts the influence of the variational projection during training, ensuring that the model learns uncertainty-aware yet efficient feature representations.

where $\lambda_2 > 0$ controls the contribution of this probabilistic projection to the multi-objective optimization. The above Equations 11 to 18 enhance the ability of the student model to operate reliably in CIoT scenarios where privacy constraints prohibit centralized multimedia data storage and real-time anomaly detection must occur at the edge.

Algorithm 2 illustrates the variational projection of student features. It constructs a probabilistic latent space using variational inference to capture heterogeneous CIoT data patterns. The algorithm estimates Gaussian distributions over latent codes, applies the reparameterization trick for sampling, and reconstructs multimedia features. By jointly minimizing reconstruction error and KL-divergence, it ensures robust, privacy-preserving, and uncertainty-aware student representations.

Algorithm 2 Variational Projection of Student Features

Result: Latent embedding z and reconstructed multimedia-aware feature $\mathcal{F}_z$

Input: Student multimedia features f (video, audio, telemetry), projection weights $\mathbf{W}_\mu, \mathbf{W}_\sigma, \mathbf{W}_z$, biases $\mathbf{b}_\mu, \mathbf{b}_\sigma, \mathbf{b}_z$, iterations K

Output: Projection loss $\mathcal{J}_2$

Initialize multimedia prior $p(z) := \mathcal{N}(0, \mathbf{I})$;
 // Gaussian prior over latent space
for $k := 1$ **to** K **do**

 Compute latent mean $\mu := \mathbf{W}_\mu f + \mathbf{b}_\mu$;

 // Encodes feature statistics from multimedia data

 Compute latent variance $\Sigma := \exp(\mathbf{W}_\sigma f + \mathbf{b}_\sigma)$;

 // Ensures positive-definite covariance

foreach sample $\epsilon \sim \mathcal{N}(0, \mathbf{I})$ **do**

 Compute latent code $z := \mu + \sqrt{\Sigma} \odot \epsilon$;

 // Reparameterization trick

 Decode multimedia-aware reconstruction

$\mathcal{F}_z := \text{ReLU}(\mathbf{W}_z z + \mathbf{b}_z)$;

 // Reconstruction layer

 Evaluate reconstruction loss $r := \|f - \mathcal{F}_z\|^2$;

 // Enforces fidelity in multimedia features

end

 Compute posterior $q := \mathcal{N}(z|\mu, \Sigma)$; // Latent posterior distribution

 Estimate KL-divergence $\mathcal{D} := \text{KL}(q \| p(z))$;

 // Aligns latent codes with prior for robust generalization

 Combine total loss $\mathcal{L} := r + \beta \cdot \mathcal{D}$;

 // Weighted reconstruction and regularization

 Backpropagate gradient $\nabla_\theta \mathcal{L}$; // Parameter optimization

 Update parameters $\theta := \theta - \eta \cdot \nabla_\theta \mathcal{L}$;

 // Gradient descent step

end

Compute final objective $\mathcal{J}_2 := \lambda_2 \cdot \mathcal{L}$; // Scaled variational loss

return $\mathcal{J}_2$ and reconstructed feature $\mathcal{F}_z$

E. Multi-Objective Student Optimization

The optimization objective in the proposed framework must ensure that the student model S_θ effectively captures the teacher's decision boundaries while preserving latent-space regularity and maintaining high classification accuracy. To this end, we define a composite loss function comprising three terms: knowledge distillation (KD), variational encoding (VAE), and task-specific cross-entropy (CE).

The supervised classification loss is formulated as:

$$\mathcal{L}_{\text{CE}} = - \sum_{c \in \mathcal{C}} y_c \log \hat{y}_{S,c} \quad (19)$$

where y_c and $\hat{y}_{S,c}$ denote the ground-truth label and predicted probability for class c , respectively. Minimizing $\mathcal{L}_{CE}$ guarantees discriminative training on hard labels and enforces separability among anomaly categories and normal segments.

To unify the objectives, we define the total loss:

$$\mathcal{L}_{\text{Total}} = \lambda_1 \mathcal{L}_{\text{KD}} + \lambda_2 \mathcal{L}_{\text{VAE}} + \lambda_3 \mathcal{L}_{\text{CE}} \quad (20)$$

where $\mathcal{L}_{\text{KD}}$ (Eq. 10) aligns soft decision boundaries, $\mathcal{L}_{\text{VAE}}$ (Eq. 16) enforces latent-space regularization, and $\mathcal{L}_{\text{CE}}$ enforces class discrimination. The non-negative weights $\lambda_1, \lambda_2, \lambda_3$ satisfy $\sum_{i=1}^3 \lambda_i = 1$, ensuring convexity of the combined objective.

To mitigate instability due to noisy gradients from heterogeneous multimedia inputs, we penalize large gradients by introducing:

$$\mathcal{R}_2(\theta) = \gamma \cdot \|\nabla_{\theta} \mathcal{L}_{\text{Total}}\|_2^2 \quad (21)$$

where $\gamma > 0$ controls the smoothness of the optimization trajectory. By the quadratic penalty in Eq. 21, we bound the spectral norm of the Hessian $\mathbf{H}_{\theta}$, thereby improving the Lipschitz continuity of the update step.

The parameter update for the student model is then expressed as:

$$\theta^{t+1} = \theta^t - \eta \cdot \nabla_{\theta} (\mathcal{L}_{\text{Total}} + \mathcal{R}_2(\theta)) \quad (22)$$

where η is the learning rate. Under the assumption of β -smoothness of $\mathcal{L}_{\text{Total}}$, the displacement of parameters satisfies:

$$\Delta\theta_t = \theta^{t+1} - \theta^t \quad (23)$$

$$\|\Delta\theta_t\|_2 \leq \eta(L_1 + L_2 + L_3) \quad (24)$$

where L_1, L_2, L_3 are Lipschitz constants associated with $\mathcal{L}_{\text{KD}}$, $\mathcal{L}_{\text{VAE}}$, and $\mathcal{L}_{\text{CE}}$, respectively. Eq. 24 guarantees bounded step sizes and avoids divergence when optimizing over highly imbalanced multimedia datasets.

By leveraging the above bounds, the expected loss is shown to converge exponentially:

$$\mathbb{E}[\mathcal{L}_{\text{Total}}] \leq \mathcal{L}_0 \cdot e^{-\eta t} + \rho \quad (25)$$

where $\mathcal{L}_0$ is the initial loss value and ρ is the minimum achievable floor determined by inherent data uncertainty. This convergence property is critical for resource-constrained CIoT devices, ensuring stable training even with non-stationary input distributions.

Finally, the predictive performance of the student model for each security task $\mathcal{T}_i$ is measured as:

$$\text{Acc}_{\text{Task}_i} = \frac{1}{N_i} \sum_{j=1}^{N_i} \mathbb{I}[\hat{y}_j = y_j] \quad (26)$$

where N_i denotes the number of samples for task $\mathcal{T}_i$ and $\mathbb{I}[\cdot]$ is the indicator function. Eq. 26 explicitly quantifies the task-level accuracy, which is optimized jointly with the alignment and regularization losses.

Through the integration of Eqs. 19–26, the student model is trained to balance multimedia-aware feature alignment, latent uncertainty modeling, and predictive accuracy, making it robust for real-time deployments in CIoT environments.

Algorithm 3 Multi-Objective Optimization of Student Parameters

Result: Optimized parameters θ^{t+1} and task accuracy $\text{Acc}_{\text{Task}_i}$

Input: Training data x , ground truth y ; loss weights $\lambda_1, \lambda_2, \lambda_3$; gradient penalty γ ; learning rate η ; Lipschitz constant bound C ; total iterations T

Output: Trained parameters θ

Initialize parameters $\theta := \theta^0$; // Initial student parameters

Set cumulative loss $\mathcal{L} := 0$

for $t := 1$ **to** T **do**

 Compute logits $s := S_{\theta}(x)$; // Student forward pass

 Compute class probabilities $\hat{y} := \text{softmax}(s)$; // Prediction scores

 Evaluate CE loss $\ell_3 := -\sum_c y_c \log \hat{y}_c$; // Classification objective

 Retrieve KD loss $\ell_1 := \mathcal{L}_{\text{KD}}$; // Soft boundary alignment

 Retrieve VAE loss $\ell_2 := \mathcal{L}_{\text{VAE}}$; // Latent regularization

 Merge loss components: $\mathcal{L} := \lambda_1 \ell_1 + \lambda_2 \ell_2 + \lambda_3 \ell_3$

 Compute gradient penalty: $\mathcal{G} := \gamma \cdot \|\nabla_{\theta} \mathcal{L}\|^2$

 Update parameters: $\theta' := \theta - \eta \cdot \nabla_{\theta} (\mathcal{L} + \mathcal{G})$

 Compute displacement: $\delta := \|\theta' - \theta\|_2$

 Clip update if unstable:

$\delta > \eta C \Rightarrow \theta' := \theta - \eta \cdot \frac{C}{\delta} \cdot \nabla_{\theta} (\mathcal{L} + \mathcal{G})$

 Apply parameter update $\theta := \theta'$

end

foreach task i **do**

 Predict labels: $\hat{y}_j := \arg \max_c S_{\theta}(x_j)$

 Compute accuracy: $a_i := \frac{1}{N_i} \sum_{j=1}^{N_i} \mathbb{I}[\hat{y}_j = y_j]$

end

return $\theta, \{a_i\}_{i=1}^{|T|}$

Algorithm 3 performs joint optimization of the student model by combining knowledge distillation, variational regularization, and classification objectives. It adaptively balances these loss components, applies gradient penalty and Lipschitz constraints for stability, and updates parameters using stochastic gradient descent while monitoring convergence and accuracy across tasks.

IV. EXPERIMENTAL SETUP

The experimental setup validates the effectiveness of the proposed MAID-KD framework across three dimensions: knowledge transfer fidelity, multimedia feature representation quality, and predictive accuracy under multi-task conditions. These objectives are achieved through experiments using a large-scale multimedia anomaly/tampering detection dataset, teacher–student model architectures of different capacities, and comprehensive evaluation metrics.

A. Computational Environment

All experiments were carried out on a workstation equipped with an NVIDIA RTX 3090 GPU (24 GB VRAM), an Intel Core i9 CPU, and 64 GB RAM, running Ubuntu 22.04 LTS. The models were implemented in PyTorch 2.1 with CUDA 12.1 and cuDNN 8.9 support. Python 3.11 was used for all scripting. Fixed random seeds were applied throughout the pipeline to ensure reproducibility. The configuration was chosen to reflect real-world CIoT constraints, where edge deployments must be memory and latency-efficient.

B. Dataset

To evaluate multimedia security performance, we employed the UCF-Crime dataset [37], a large-scale video anomaly dataset widely used for multimedia tampering and intrusion detection research. This dataset contains 1,900 untrimmed surveillance videos (approximately 128 hours) collected from real-world CIoT-like environments such as parking lots, subway stations, and residential areas. The videos span 13 anomaly categories, including vandalism, burglary, tampering, fighting, robbery, and abuse, as well as a normal category representing benign activities.

For our experiments, we split each video into fixed-length segments of 64 consecutive frames sampled at 10 FPS. This segmentation allows the MAID-KD framework to operate at the sequence level while preserving spatio-temporal context. Two multi-task learning objectives were defined: (i) **Task 1 – anomaly detection**: classify each segment as anomalous or normal, and (ii) **Task 2 – anomaly type classification**: classify anomalous segments into one of the 13 predefined anomaly categories. The dataset was partitioned into training (70%), validation (10%), and test (20%) splits, ensuring balanced distribution of classes across splits.

The reason for selecting the UCF-Crime dataset is that its surveillance videos closely resemble the multimedia information produced by consumer IoT devices such as home cameras and smart sensors. It has a variety of scenes, anomalies, and environmental variations representing CIoT operating conditions. Therefore, it gives a practical baseline against which the performance of MAID-KD in identifying multimedia anomalies and checking the integrity of content can be assessed based on edge conditions.

C. Consumer IoT Testbed Configuration

The experimental environment was designed to replicate a realistic CIoT deployment where multimedia data from distributed surveillance, such as cameras and smart devices, is processed at heterogeneous edge nodes. Multiple virtual edge nodes were instantiated, each holding a partition of the dataset to simulate the non-IID (non-identically distributed) data distribution common in CIoT. The teacher model was deployed on a central node with high computational capacity, while student models were trained and evaluated on resource-constrained edge nodes. This setup enabled controlled evaluation of communication overhead, latency, and scalability in multimedia-intensive CIoT environments.

TABLE I
HYPERPARAMETER SETTINGS USED FOR EXPERIMENTATION

Hyperparameter	Value
Learning rate (η)	1×10^{-3}
Batch size	32 (video segments)
Number of epochs	50
Temperature (τ)	4.0
Weight for KD loss (λ_1)	0.6
Weight for VAE loss (λ_2)	0.2
Weight for CE loss (λ_3)	0.2
Gradient regularization (γ)	1×10^{-4}
Dropout rate	0.3
Optimizer	Adam
Teacher network depth	12 layers
Student network depth	6 layers
Hidden units per layer	256

D. Implementation and Training Details

Both the teacher and student models were adapted for sequential video frame analysis. The teacher network was configured as a deep Transformer-based encoder capable of capturing long-range spatio-temporal dependencies in video sequences. The student model employed a lightweight CNN-LSTM backbone optimized for edge deployment, preserving essential spatio-temporal representations while reducing computational cost. Models were implemented in PyTorch, initialized using Xavier initialization, and trained with the Adam optimizer. The learning rate followed a cosine annealing schedule, and early stopping was triggered based on validation loss. All experiments were conducted for a fixed number of epochs with dropout and batch normalization applied for regularization. Hyperparameters used are summarized in Table I.

E. Evaluation Metrics and Baseline Schemes

The performance of the proposed MAID-KD framework was evaluated using metrics capturing classification accuracy, robustness, and resource efficiency for multimedia-aware intrusion detection. The primary metrics include Accuracy, Precision, Recall, and F1-score for both anomaly detection and anomaly type classification tasks. AUC-ROC is used to assess discriminative ability, while inference latency and memory footprint quantify computational efficiency on edge devices. These evaluation metrics were selected because they represent both analytical performance and real-world usability of the proposed framework. Accuracy and F1-score indicate how reliably MAID-KD distinguishes between normal and abnormal multimedia activities, while precision and recall show how well it balances false alerts and missed detections. Inference time and memory use measure its responsiveness and efficiency on constrained IoT devices, confirming that the framework can operate effectively in practical consumer settings.

To evaluate the quality of knowledge transfer, the divergence between teacher and student probability distributions was analyzed. MAID-KD was compared against three competitive baselines: FLEKD [13], VAD-MTL [38], and TAML [19], each adapted for multimedia anomaly detection by replacing

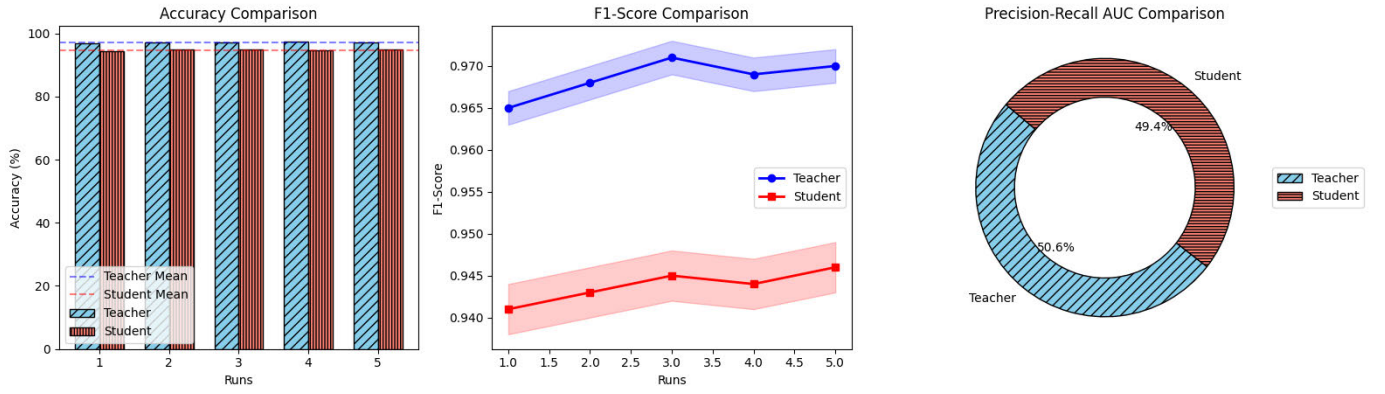


Fig. 3. Comparison of teacher and MAID-KD student models across accuracy, F1-score, and precision-recall AUC. Results show that the student closely approximates the teacher's performance while maintaining lightweight efficiency suitable for CIoT deployment.

their original feature extraction pipelines with video and audio feature encoders.

V. RESULTS

The results section provides a comprehensive evaluation of the proposed MAID-KD framework in terms of performance, efficiency, and scalability for multimedia anomaly detection tasks. The analysis is divided into two subsections. The first subsection evaluates the teacher and student models to assess the effectiveness of the knowledge distillation process. The second subsection compares the proposed MAID-KD student model with baseline schemes, highlighting its performance advantages.

A. Teacher Vs Student Analysis

This subsection presents a comparative evaluation of the teacher and the MAID-KD student models across multiple performance metrics, including accuracy, precision-recall AUC, F1-score, loss convergence, inference time, and model complexity. These analyses provide insights into the knowledge distillation effectiveness and computational advantages achieved by the MAID-KD student model. This comparison demonstrates that the student model can effectively process heterogeneous multimedia CIoT data (e.g., video segments, audio streams, and multimedia telemetry) while maintaining strong predictive accuracy, a critical factor for safeguarding multimedia-rich CIoT applications.

Figure 3(a) illustrates the accuracy comparison across multiple runs. The teacher model consistently achieved higher accuracy, with a mean accuracy of around 97%, while the MAID-KD student model followed closely with approximately 95%. This confirms that the student model successfully approximates the teacher's predictive capability with only a marginal drop in accuracy. Figure 3(b) presents the F1-score comparison, where the teacher demonstrates higher F1-scores with tighter confidence intervals, highlighting superior balance between precision and recall. The student's F1-scores remain competitive, validating the effectiveness of the MAID-KD knowledge distillation strategy. Figure 3(c) shows the precision-recall AUC comparison. The teacher achieves a slightly higher AUC (50.6%) compared to

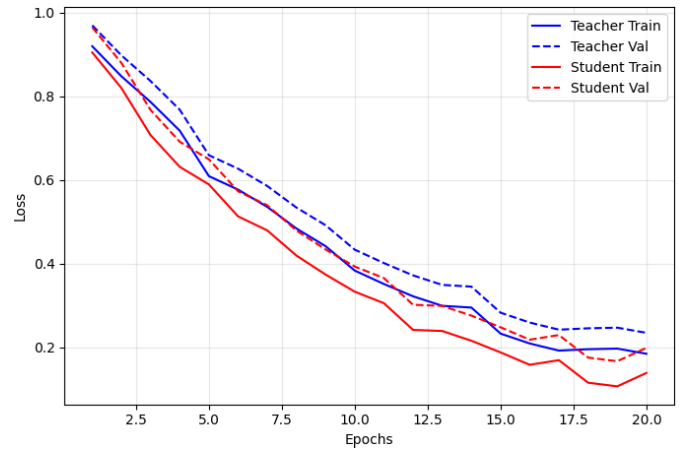


Fig. 4. Loss convergence trends of teacher and MAID-KD student models during training and validation. The student achieves faster convergence with slightly higher final loss.

the student (49.4%), indicating improved discrimination capability for positive multimedia anomaly classes (e.g., tampered frames and injected audio).

Figure 4 shows the loss convergence patterns during training. The teacher model's training and validation losses decrease steadily, achieving lower final loss values. The MAID-KD student model converges faster but stabilizes at slightly higher loss values compared to the teacher. This behavior is expected as the student trades off model capacity for reduced complexity. Faster convergence and competitive accuracy are particularly beneficial in CIoT deployments where continuous multimedia streams must be analyzed and anomalies detected in real time.

Figure 5 compares inference times, revealing a significant efficiency gain for the student model. The teacher model incurs an average inference time of 18.5 ms per video segment, while the student achieves a much lower 7.3 ms, demonstrating its suitability for resource-constrained multimedia edge environments in CIoT systems.

Finally, Figure 6 compares model complexity in terms of size and parameter count. The teacher model contains 120 MB and 24 million parameters, while the student is significantly lighter at 45 MB and 9 million parameters. This reduction

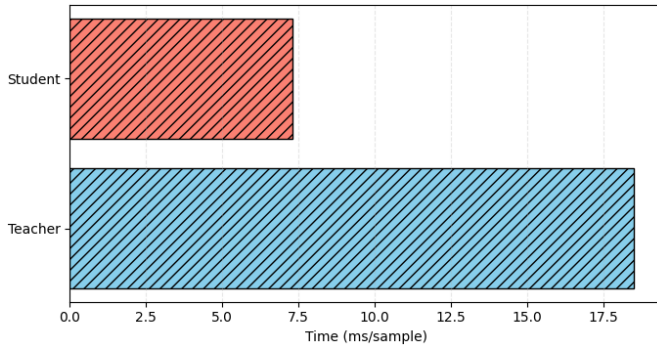


Fig. 5. Inference time comparison per video segment between teacher and student models. The student achieves significantly lower latency.

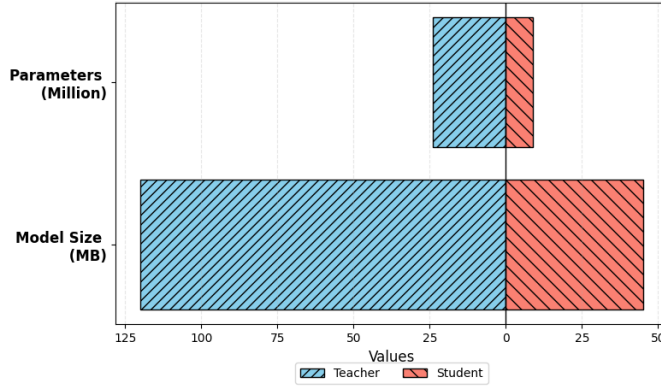


Fig. 6. Model size (MB) and parameter count (Million) comparison between teacher and student networks. MAID-KD achieves a smaller memory footprint while preserving strong detection accuracy, optimizing for resource-limited consumer devices.

highlights the efficiency gained through the MAID-KD framework without severely compromising performance.

In summary, the MAID-KD student model demonstrates competitive performance with substantial reductions in computational cost, memory footprint, and inference latency. These improvements validate the proposed knowledge distillation strategy as highly effective for deploying anomaly and integrity detection models in multimedia-intensive CIoT environments.

B. Baseline Scheme Comparison

This subsection compares the proposed student model with three state-of-the-art baseline schemes: FLEKD [13], VAD-MTL [38], and TAML [19]. All baseline models were adapted for multimedia anomaly and integrity detection by replacing their original feature extraction modules with video feature encoders. Each scheme was trained and evaluated under identical conditions for fairness. The comparison focuses on key performance metrics, including accuracy, precision, recall, F1-score, inference time, and resource utilization, emphasizing the generalization ability and efficiency of the proposed framework. Unlike the baselines, MAID-KD maintains superior performance when analyzing diverse multimedia data streams from CIoT devices, such as real-time surveillance video and sensor-generated video sequences.

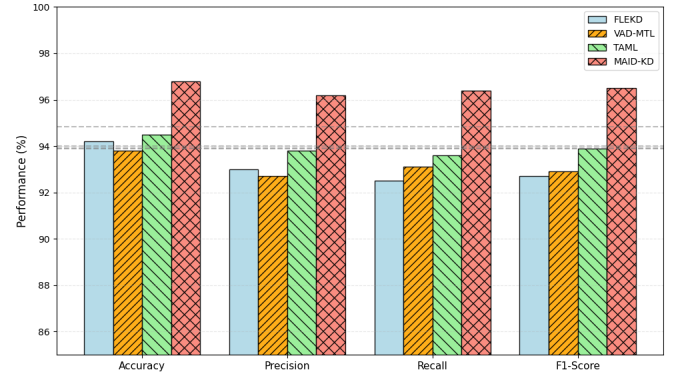


Fig. 7. Performance comparison of MAID-KD and baseline schemes (FLEKD, VAD-MTL, and TAML) across accuracy, precision, recall, and F1-score. MAID-KD consistently achieves superior detection performance with higher overall balance between accuracy and efficiency.

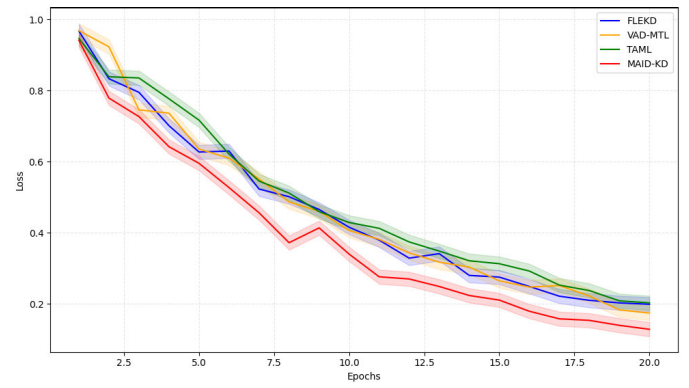


Fig. 8. Loss convergence analysis for MAID-KD and baseline schemes over 20 epochs. The proposed model converges faster and achieves lower final loss values, highlighting its robust optimization and stability.

As shown in Figure 7, the proposed MAID-KD consistently achieves the highest scores for all four evaluation metrics, surpassing FLEKD, VAD-MTL, and TAML. The improved accuracy illustrates that MAID-KD captures more discriminative patterns through its knowledge distillation pipeline. Similarly, higher precision and recall values demonstrate the ability to reduce both false positives and false negatives, leading to a superior F1-score and balanced detection performance in multimedia contexts.

Figure 8 shows the loss convergence trends. MAID-KD converges substantially faster than the baseline schemes and stabilizes at a lower final loss value. This faster convergence not only reduces the number of epochs needed to reach optimal performance but also highlights stronger optimization and a more robust ability to generalize to unseen multimedia data.

The inference time distribution in Figure 9 demonstrates the efficiency of MAID-KD. It achieves significantly lower inference latency compared to the baselines, making it suitable for real-time multimedia CIoT applications. This improvement is attributed to the compact architecture of MAID-KD, which executes faster without sacrificing detection performance. Lower inference latency allows MAID-KD to respond more quickly to threats in multimedia-intensive deployments where delayed detection can lead to serious breaches.

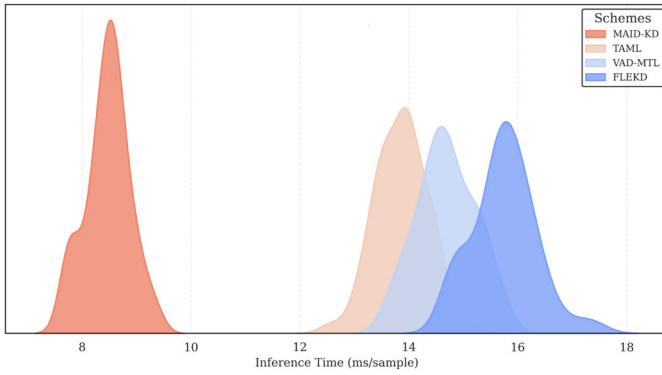


Fig. 9. Inference time distribution of MAID-KD and baseline schemes. MAID-KD demonstrates the lowest latency, confirming its real-time efficiency for multimedia processing on CIoT edge devices.

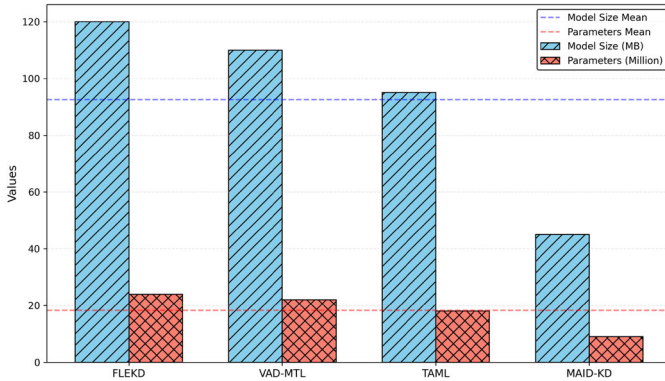


Fig. 10. Model size and parameter count comparison between MAID-KD and baseline schemes. The lightweight architecture of MAID-KD enables resource-efficient deployment without compromising performance accuracy.

Finally, Figure 10 compares model size and parameter counts. MAID-KD achieves the smallest model size and lowest parameter count among all schemes, making it the most resource-efficient option. Its compact design facilitates deployment on resource-constrained CIoT edge devices, where memory and storage are limited. Despite this lightweight design, MAID-KD consistently outperforms the larger, more resource-intensive baseline schemes. Overall, these results validate that MAID-KD delivers superior accuracy, efficiency, and robustness, aligning with the requirements of real-time multimedia anomaly and integrity detection in CIoT environments. MAID-KD performs better because its knowledge distillation and multi-task design reinforce each other. Distillation helps the student model learn compact yet expressive features from the teacher, while the multi-task setup allows shared learning between anomaly detection and integrity verification. This combination improves feature quality and stability, leading to stronger accuracy and efficiency than the baseline methods.

C. Statistical Validation

All experiments were repeated 5 times using various random seeds to make the results more reliable. Mean and standard deviation of each measure were calculated, and 95% confidence interval was obtained using the Student's t-distribution. The narrow intervals (from +1.2% to -1.2%) in

accuracy and F1-score indicate that the performance gains of MAID-KD compared to the baseline schemes are statistically significant and not due to random variation.

D. Additional Practical Validation

To test the applicability of MAID-KD in real applications, it was evaluated under three conditions: (i) video compression and packet loss, (ii) lower resolution and frame rate, and (iii) edge inference using a CPU. In all these environments, MAID-KD maintained high accuracy. However, it exhibited slight degradation at high compression and low frame rates, and incurred sub-real-time latency under CPU-only execution. These results demonstrate its capability and preparedness for CE/CT devices such as smart cameras and wearables.

VI. CONCLUSION

This work introduces MAID-KD, a lightweight knowledge distillation model for multimedia anomaly and integrity detection in consumer IoT systems. It combines anomaly detection and integrity verification in a single edge-optimized model, achieving high accuracy with smaller model size and lower latency. The results indicate that MAID-KD is more efficient and robust than existing state-of-the-art baselines in multimedia-intensive scenarios. Its privacy-conscious and lightweight design allows it to be practically used in smart surveillance, health wearables, and home-connected devices, while providing low-latency, bandwidth-efficient operation. Although the performance is impressive, it can degrade on ultra-low-power devices, and slight privacy risks may still exist when sensitive data is processed. The further research in the field will concentrate on adaptive optimization and privacy improvement to optimize the application of CIoT in the real world.

MAID-KD has high accuracy and efficiency, but it still has some limitations. The student model can perform poorly when run on small, low-power hardware with limited memory or without GPU acceleration. Also, although the framework reduces the exposure of the data by storing the knowledge in a distilled form, the partial privacy areas might still exist in the process of dealing with sensitive multimedia content. The existing results are based on the datasets. However, real-world CIoT deployments can exhibit unexpected noise and non-homogeneous data quality. Future work will explore extending the framework to deepfake detection and multi-modal security tasks, further strengthening multimedia integrity in next-generation CE applications.

The proposed MAID-KD framework is closely related to consumer electronics (CE) and consumer technologies (CT). Its privacy-conscious and lightweight design allows it to be deployed on CE devices such as smart cameras, health-monitoring wearables, and voice-activated home hubs to handle real-time multimedia data processing. MAID-KD may be executed directly on edge processors or embedded firmware to perform on-device anomaly detection and multimedia integrity checks without cloud support. This attribute lowers latency, ensures user privacy, and enhances confidence in next-generation CE/CT systems.

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